

8.1.1 Electrostatic force:

In the early days, the early Greek philosophers have studied the physics of electrostatic by performing an activity; when amber is rubbed with silk or wool and brought near dust particles, the dust particles will jump to stick with the amber. This attraction between dust particles and amber is known as an electrostatic force.

The electrostatic force between two charges is analogous to the gravitational force. Two charged objects can (i) attract or (ii) repel each other with a force known as electrostatic force.

Whereas the gravitational force is always attractive in nature. Furthermore, the electrostatic forces are short range forces as compared to the gravitational forces, but the electrostatic forces are stronger than the gravitational forces.

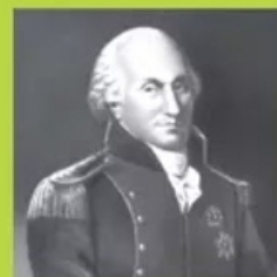
The object having the same nature of charges repels each other and having opposite nature attracts each other with an electrostatic force.

8.1.2 Coulomb's Law:

Charles-Augustin de Coulomb performed an experiment of torsion balance (see figure 8.1) in 1785 to measure the magnitudes of the electric force between charged objects. Once the spheres are charged by rubbing, the electric force between them is very large compared with the gravitational attraction, and so the gravitational force can be neglected. From Coulomb's experiments, we can generalize the properties of the electric force (sometimes called the electrostatic force) between two stationary charged particles.

- If two charged particles are brought near each other, they exert an electrostatic force on each other. The direction of the force vectors depends on the signs of the charges.
- If the particles have the same sign of charge, they repel each other. That means that the force vector on each is directly away from the other particle. Besides this,

DO YOU KNOW?



- Name: Charles-Augustin de Coulomb
- Birth Year: 1736
- Birth date: June 14, 1736
- Birth City: Angoulême
- Birth Country: France
- Gender: Male
- Best Known For: French engineer and physicist Charles de Coulomb made pioneering discoveries in electricity and magnetism, and came up with the theory called Coulomb's Law.

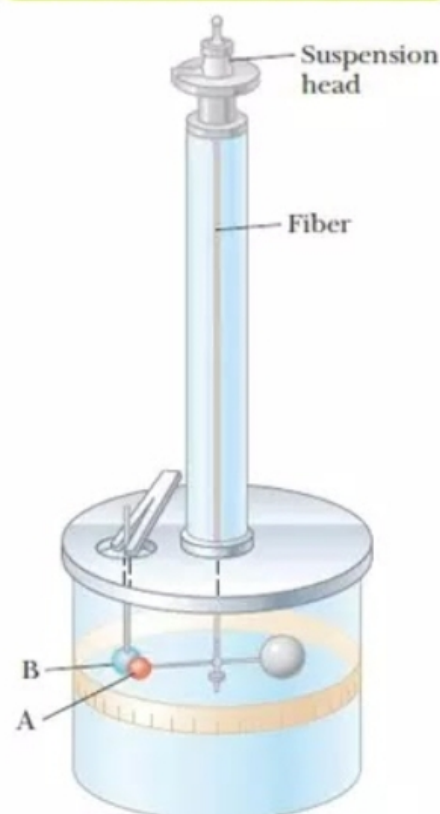


Fig: 8.1
Coulomb's torsion balance.

- If the particles have opposite signs of charge, they attract each other and the force vector on each is directly towards the other particle as shown in figure 8.2.

The illustration for two charges (q_1, q_2) and separation (r) between the charges is shown in figure 8.3.

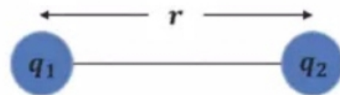


Fig: 8.3
Two charges q_1 and q_2 placed at a distance r from each other.

The electrostatic force F is directly proportional to the product of magnitudes of the charges q_1 and q_2 . And inversely proportional to the square of the distance r between the charges and

$$\vec{F} \propto q_1 q_2$$

$$\vec{F} \propto \frac{1}{r^2}$$

This result is known as the inverse square law.

$$\vec{F} = k \frac{q_1 q_2}{r^2}$$

Where k is known as the Coulomb's constant and $k = \frac{1}{4\pi\epsilon_0}$.

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad (8.1)$$

8.1.3 Coulomb's force in different medium:

The permittivity of air at standard pressure is only about 1.005 times that of ϵ_0 . Therefore, it can be assumed in many cases that the values of ϵ_0 are equal for both vacuum and air. The permittivity of water is about eighty times that of a vacuum. Thus, the force between charges situated in water is eighty times less than if they were situated the same distance apart in a vacuum.

In general, the equation (8.1) can rewrite as

$$\vec{F} = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2}$$

where $\epsilon = \epsilon_r \epsilon_0$ is the permittivity of the medium and ϵ_r is the relative permittivity.



Fig: 8.2 (a) both are positive

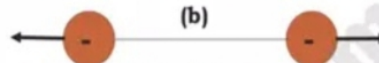


Fig: 8.2 (b) both are negative..

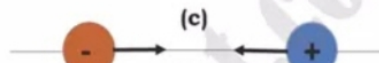


Fig: 8.2 (c) both are opposite

DO YOU KNOW?

For your information

ϵ_0 is known as the permittivity of free space. if we suppose the charges are situated in a vacuum then the value of ϵ_0 is given as $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N m}^2$. The value of Coulomb's constant in SI unit is given as $k = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2$

Material	Relative Permittivity
Vacuum	1.0000
Air	1.0006
PTFE, FEP (Teflon)	2.0
Polypropylene	2.20 to 2.28
Polystyrene	2.4 to 3.2
Wood (Oak)	3.3
Bakelite	3.5 to 6.0
Wood (Maple)	4.4
Glass	4.9 to 7.5
Wood (Birch)	5.2
Glass-Bonded Mica	6.3 to 9.3
Porcelain, Steatite	6.5

DO YOU KNOW?

Why do salt crystals dissolve in water?

A salt crystal is an ionic crystal. For example, in the case of sodium chloride (NaCl). The sodium ions (Na^+) and the chlorine ions (Cl^-) in a salt crystal are oppositely charged. The electrostatic forces between these ions hold the atoms together.. Note that Coulomb's law as stated above applies strictly to a vacuum. In terms of the force between two-point charges in air, the force is effectively the same as it would be in a vacuum.

Vector form of Coulomb's law:

Figure 8.4 illustrates the vector form of electrostatic force between two charges. The electric force F_{12} exerted by a charge q_1 on other charge q_2 is written as:

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2} \hat{r}_{12} \quad (8.2)$$

where $\hat{r}_{12}$ is a unit vector directed from q_1 toward q_2 as shown Figure 8.4. Because the electric force obeys Newton's third law, as equal and opposite direction; that is, $\vec{F}_{12} = -\vec{F}_{21}$

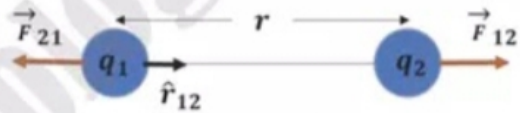


Fig: 8.4

The force F_{21} exerted by q_2 on q_1 is equal in magnitude and opposite in direction to the force F_{12} exerted by q_1 on q_2 .

Worked Example 8.1

The electron and proton of a hydrogen atom (figure A) are separated by approximately $5.3 \times 10^{-11} \text{ m}$. Find the magnitudes of (a) the electric force and (b) the gravitational force between the two particles. (c) What is your conclusion about these forces.

Solution:

Step:1 Write down the known quantities and quantities to be found.

$$r = 5.3 \times 10^{-11} \text{ m}.$$

- (a) Charge on electron = $q_e = -1.602 \times 10^{-19} \text{ C}$,
 Charge on proton = $q_p = 1.602 \times 10^{-19} \text{ C}$,
 $k = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2$
 Electrostatic force = $F_e = ?$

- (b) Mass of electron = $m_e = 9.109 \times 10^{-31} \text{ kg}$
 Mass of proton = $m_p = 1.672 \times 10^{-27} \text{ kg}$.
 where $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$,
 Gravitational force = $F_g = ?$

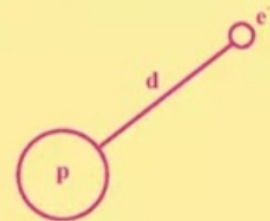


Figure A

Unit 8: Electric Fields

Step:2 Write down the formula and rearrange if necessary.

(a) For electrostatic force $F_e = k \frac{q_e q_p}{r^2}$

(b) For gravitational force $F_g = G \frac{m_e m_p}{r^2}$

Step:3 Put the values in formula and calculate.

$$(a) F_e = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2 \times \frac{-1.602 \times 10^{-19} \text{ C} \times 1.602 \times 10^{-19} \text{ C}}{(5.3 \times 10^{-11} \text{ m})^2}$$

$$F_e = -8.2 \times 10^{-8} \text{ N}$$

where negative sign indicates the force is attractive. The magnitude of this electrostatic force is $8.2 \times 10^{-8} \text{ N}$.

$$(b) F_g = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2 \times \frac{9.109 \times 10^{-31} \text{ kg} \times 1.672 \times 10^{-27} \text{ kg}}{(5.3 \times 10^{-11} \text{ m})^2}$$

$$F_g = 3.6 \times 10^{-47} \text{ N}$$

Hence, the gravitational force of attraction between the particles is $3.6 \times 10^{-47} \text{ N}$

(c) Conclusion:

The gravitational force between electron and proton is negligible as compared to the electric force, which implies that electric force is strong force.

Self-Assessment Questions:

1. Calculate the separation between two electrons (in vacuum) for which the electric force between them is equal to the gravitation force on one of them at the earth surface.
2. If a dielectric medium of dielectric constant ϵ_r is filled in between the charges which are placed at distance r , then show that force between the charges decreases as compare the force when these charge place in air at same distance.

8.2.1 Electric field:

The concept of a field was developed by Michael Faraday (1791–1867) in the context of electric forces.

an electric field is said to exist in the region of space around a charged object. The charged object which produce the electric field is known as the source charge. Michael Faraday also introduced the idea of the electric field lines. These electric lines are imaginary, but these lines can be visualized by the motion of the test charge. The electric field lines around charged objects are very helpful in understanding the field strength and direction of electric field. These field lines are radial and cannot intersect each other. These field lines are originated from a positive charge and terminated on a negative charge as shown in figure 8.5. The electric field is strong where these lines are close together.



Fig: 8.5

The electric lines of force for positive and negative charge.

Strength of Electric field:

To observe the strength of electric field of the source charge, another charged object is required which is known as the test charge. This test charge experiences the electric force due to the field produced by the source charge.

To be more specific, the electric field $\vec{E}$ at a point in space around the source charge is defined as the electric force F acting on a positive test charge placed at that point divided by the test charge q_0 .

$$\vec{E} = \frac{\vec{F}}{q_0} \quad (8.3)$$

The SI unit of electric field is volts per meter (V/m) or, equivalently, newtons per coulomb (N/C).

DO YOU KNOW?

To eliminate stray electric field interference, circuits of sensitive electronic devices such as T.V and computers are often enclosed with metal boxes.



8.2.2 Electric field of point charge:

To find the electric field due to a charged particle, consider a point charge q as a source charge. This charge creates an electric field at all points in space surrounding it. A test charge q_0 is placed at a distance r from the source charge. (see figure 8.6).

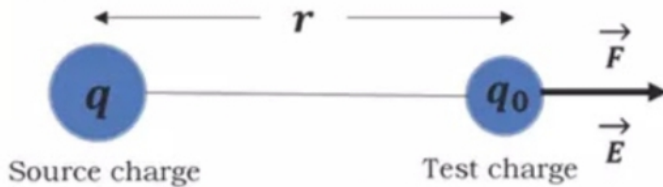


Fig: 8.6

A test charge q_0 in the field of source charge q .

The test charge experiences electric force from which we can determine the direction of the electric field. According to Coulomb's law, the force exerted by q on the test charge is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r}$$

where $\hat{r}$ is a unit vector directed from q toward q_0 and the direction of force $\vec{F}$ is directly away from the source charge q . To calculate the electric field at a distance r from the point charges, we first calculate the electric field vector using equation 8.3.

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (8.4)$$

The direction of electric field directly away from the point charge q as shown in figure 8.9.

The magnitude of electric field at any given distance r can be expressed as:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\vec{E} \propto \frac{1}{r^2}$$

This equation shows that the intensity of the electric field decreases with distance from the source charge(s). The field intensity is inversely proportional to the square of the distance from the charge(s).

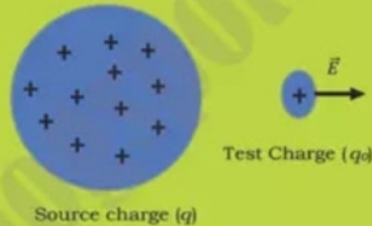
If there are n number of charged particles $q_1, q_2, q_3, \dots, q_n$, then the net electric force of these charges at a point. According to superposition principle,

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n$$

From equation 8.6, the net electric field of these charged particles

DO YOU KNOW?

A small positive test charge q_0 placed near a source charge carrying a much greater positive charges q as shown in figure. The test charge is always so small that the field of the source charge is unaffected by its presence.



The electric field strength due to the source charge at the location of the test charge is defined as the electric force on the test charge per unit charge.

$$\vec{E} = \frac{\vec{F}_1}{q_0} + \frac{\vec{F}_2}{q_0} + \frac{\vec{F}_3}{q_0} + \dots + \frac{\vec{F}_n}{q_0}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots + \vec{E}_n$$

Therefore, the electric fields of these charges experienced by the test charge q_0 is vector sum of electric field of individual charge.